

Ioana Popescu

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EDUCATION

ETH Zurich

Master in Data Science

Sep 2025 – Jun 2027 (expected)
Zurich, Switzerland

- Current GPA: 6.0/6.0
- Recipient of ESOP Excellence Scholarship — ETH Foundation, Sep 2025

École Polytechnique

Bachelor in Mathematics and Computer Science

Sep 2022 – Jun 2025
Palaiseau, France

- Exchange Semester — ETH Zurich, Department of Mathematics — Grade: 6.0/6.0
- GPA: 4.29/4.30 — Valedictorian
- Recipient of Excellence Scholarship — Fondation de l'X, Sep 2022

AWARDS AND HONORS

Mathematical Olympiads

- Romanian Mathematical Olympiad (RMO): Gold Medal (ranked 1st, maximum score), Apr 2022
- European Girls' Mathematical Olympiad (EGMO): Gold Medal (ranked 5th in Europe), Apr 2021

EXPERIENCE

Quantitative Research Intern — Optiver

Jul 2026 – Sep 2026

- Traded Euro Stoxx index in a simulated market and conducted a research project identifying profitable options trading strategies using seven years of S&P 500 and Nasdaq market data (skew, calendar swaps, vol mean-reversion)

Statistics Research Intern — University of Cambridge

Jul 2025 – Aug 2025

- Research in statistics on expressivity of transformers for in-context learning, supervised by Prof. Richard Samworth. We showed that a pretrained transformer with small number of parameters and pretraining-sequences is minimax-optimal for ICL regression problems. Accepted at ICML 2026: <https://arxiv.org/abs/2601.15014v1>

Software Engineer Intern — Microsoft

Jul 2024 – Sep 2024

- Improved the design of the Office products checkout website (React, TypeScript, CSS). Ran an A/B test adding tooltips to Microsoft products, deployed to millions of users to enhance UX.

Data Science Intern — Tenaris

Jun 2023 – Jul 2023

- Built a CNN to monitor steel pipe piercing, balancing execution time and recall in Python. Applied data augmentation, hyperparameter tuning (Weights & Biases) and visualization techniques (heatmaps). Evaluated sliding window, one-shot, and YOLO approaches; deployed the best-performing model as a web service

ACADEMIC PROJECTS

Research Project: Reconstruction of 3D Spatial Transcriptomics Volumes

Feb 2026 – Jun 2026

Developed a generative model to reconstruct missing slices in 3D spatial transcriptomics volumes, via optimal transport and thin plate splines and benchmarked against existing methods; supervised by Prof. Valentina Boeva (ETH Zurich)

Coding Project: Single-cell RNA-seq cell type clustering

Nov 2025 – Dec 2025

Worked with scRNA-seq data, applying normalization, dimensionality reduction, and batch integration (Harmony, Seurat, scVI). Benchmarked unsupervised (Leiden/Louvain) and semi-supervised (logistic regression) clustering methods to optimize cell-type classification, as part of the ML for Genomics course at ETH Zurich.

Bachelor Thesis: Energy of Random Trees

Jan 2025 – Mar 2025

Researched the asymptotic energy of preferential attachment trees using probability theory, supervised by Prof. Igor Kortchemski (ENS Ulm); received special mention from the jury of “Junior Maryam Mirzakhani Prize 2025”

Coding Project: Evolution Simulator

Nov 2023 – Jan 2024

Developed an evolutionary simulation using the NEAT algorithm, where neural network-based agents evolve complex behaviors without hardcoded rules. Code: github.com/EvolutionSimulator/EvolutionSimulator